HR-ETE-GNN: Robust Volatility Forecasting via Hurst-Regime-Adaptive Rényi Effective Transfer Entropy Graph Neural Networks

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ABSTRACT

Accurate volatility forecasting is critical for portfolio optimization, risk management, and derivatives pricing. Recent work by Lee and Cho (2025) demonstrated that Hurst Exponent Effective Transfer Entropy Graph Neural Network (H-ETE-GNN) which constructs directed inter-asset graphs from Shannon-based Effective Transfer Entropy (ETE) and adapts to regime changes via the Hurst exponent outperforms traditional univariate and correlation-based models for one-day-ahead realized volatility prediction. However, the authors acknowledged that their model's performance deteriorates during extreme market conditions, attributing this to Shannon entropy's equal weighting of all regions of the return distribution, which dilutes tail-event information that dominates crisis periods. This paper addresses that limitation by replacing the Shannon-based ETE with Effective Rényi Transfer Entropy (ER-TE) which is a generalization grounded in the theoretical framework of Jizba, Lavička, and Tabachová (2022) where a tunable order parameter a allows the measure to emphasize rare, extreme events ($\alpha < 1$) or typical events ($\alpha > 1$), reducing to Shannon TE at $\alpha -> 1$. We denote the resulting architecture **HR-ETE-GNN** (Hurst + Rényi Effective Transfer Entropy Graph Neural Network). Using daily price data of 10 major country-level ETFs over 19 years (2003-2024), Forecasting performance is assessed via QLIKE as the primary metric for comparing model performance.

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CHAPTER I

INTRODUCTION

1.1 Context of the Study

Volatility prediction in financial markets is important when it comes to risk management [1, 2]. Volatility refers to the magnitude of fluctuations in an asset's returns over a given time horizon. Formally, it is most commonly operationalized as the standard deviation of asset returns and serves as the measure of financial risk in modern asset pricing, and portfolio construction [1,2]. In recent years, different models were employed to forecast volatility ranging from traditional models like Generalized Autoregressive Conditional Heteroskedasticity (GARCH) [3] to deep learning models like Long Short-Term Memory (LSTM) [4]. Among the latter, Graph Neural Networks (GNNs) have been used as a model for financial time series because of their capabilities to model interdependencies among assets in a graph, where nodes represent individual securities and edges encode the strength and direction of inter-asset relationships [5].

The graph tells the network which assets influence which. Granger Causality introduces directionality but assumes linearity and stationarity [6]. Transfer Entropy (TE), introduced by Schreiber [7], offers a model-free, non-parametric measure of directional information flow that captures both linear and non-linear dependencies, making it theoretically superior for financial data characterized by fat tails, volatility clustering, and asymmetric dependence structure [8].

Lee and Cho [9] proposed a hybrid model that further advances the proposed models with the H-ETE-GNN model which integrates three components: (1) Effective Transfer Entropy that corrects a positive bias that Transfer Entropy creates introduced

by Marschinski and Kantz [10]. This was used to quantify information flow among ten country exchange-traded funds (ETFs). (2) A Graph Convolutional Network (GCN). It is then used to construct graph convolutional networks utilizing ETE-based graphs to forecast one-day-ahead realized volatility. (3) A Hurst-exponent-based regime detection mechanism that retrains the graph when the market transitions between persistent and anti-persistent states. Their empirical results over a 19-year sample (2003–2024) demonstrated significant improvements over LSTM, GRU, and GNN baselines using Granger Causality, Pearson Correlation, or standard Transfer Entropy as edge metrics.

However, Lee and Cho [9] identified a limitation with their hybrid model. The model's performance degraded during extreme market events. They attributed this degradation to two factors: the Hurst exponent operating under a monofractal assumption, which may not fully capture the asymmetric and complex nature of the market structure; and the limitation of ETE in detecting extreme, localized nonlinear interactions, potentially missing crucial regime changes. This has led them to recommending algorithms that can detect event significance or extreme probabilities. One of their recommendations to incorporate extreme probabilities is Rényi-entropy-based Transfer Entropy to better capture key information flow, increasing its robustness and adaptability across diverse market regimes, including crises.

1.2 Research Objectives

The objective of this research is to develop and evaluate HR-ETE-GNN, a Hurst regime-adaptive graph neural network that employs Effective Rényi Transfer Entropy (ER-TE) in place of Shannon-based Transfer Entropy for one-day-ahead realized-volatility (RV) forecasting across country-level iShares MSCI ETFs, with the aim of improving forecast robustness under extreme market regimes while preserving calm-period performance. This study operationalizes Lee and Cho's [9] recommendations of their H-ETE-GNN hybrid model and the theoretical results of Rényi Transfer Entropy of Jizba, Lavička, and Tabachová [11].

In support of this, the specific objectives are as follows:

1. Design and implement an ER-TE based adjacency construction using Rényi entropy with shuffled-surrogate bias correction that substitutes Shannon ETE in the H-ETE-GNN pipeline.
2. To evaluate whether HR-ETE-GNN based graph construction yields statistically significant reduction in expected forecast loss relative to H-ETE-GNN
3. To compare ER-TE and Shannon ETE network topologies across crisis and non-crisis windows, identifying whether tail-sensitive information flow changes which countries are dominant transmitters of volatility information.

1.3 Research Questions

1 Does replacing Shannon-Based Effective Transfer Entropy (ETE) with Effective Rényi Transfer Entropy (ER-TE) as the graph construction metric in the Graph Neural Network of the H-ETE-GNN framework:

(a) Produce a statistically significant reduction in expected QLIKE loss for one-day-ahead realized-volatility forecasts relative to H-ETE-GNN

2 When the directed weighted adjacency matrix is reconstructed from ER-TE versus Shannon ETE on the same crisis and non-crisis windows;

(a) Does the resulting topology differ in a measurable way – specifically, do the **top volatility information transmitters change** identity under tail sensitive $\alpha < 1$ measurement during crisis periods relative to Shannon measurement

1.4 Scope and Limitations

Scope

The analysis is restricted to ten country-level iShares MSCI ETFs (EWA, EWC, EWG, EWJ, EWT, EWU, EWW, EWY, EWZ, and EZA), observed at daily frequency from February 2003 through May 2024. The MSCI World ETF is used exclusively as a global regime proxy for Hurst-based detection and is not included in the forecasting target.

In advancing the H-ETE-GNN hybrid model, only the entropy basis used for graph construction is changed. The multi-scale 1D convolution, three GCN layers, MSE loss, Adam optimizer, and early-stopping criteria are held identical to Lee and Cho [9]. ER-TE is estimated with $k = 1$, an α-grid of $\{0.3, 0.5, 0.7, 0.9, 1.0, 1.5, 2.0\}$, and m-shuffled surrogates for bias correction. Comparisons are drawn against ETE-GNN, TE-GNN, Granger-GNN, Pearson-GNN, GRU, LSTM, and H-ETE-GNN as the primary baseline.

The study does not consider intraday data, alternative realized-volatility measures, sector or single-stock universes, attention-based architectures such as GAT, or adaptive/regime-contingent α selection.

Limitations.

1. A vanilla GCN without attention is retained to preserve cleanliness, and a GAT-style mechanism is deferred to avoid confounding attribution.
2. The k-NN Rényi Transfer estimator exhibits higher variance at extreme α values and in higher dimensions; results at $\alpha = 0.3$ fall below the theoretical α-Gaussian validity lower bound of $\alpha > 0.4$ for $k = 1$ and are reported with caution.
3. The use of a single global regime proxy (MSCI World) presumes a synchronized regime shift across all countries and may fail to capture regional differences.
4. Comparisons against the secondary benchmarks are descriptive. The results do not establish that HR-ETE-GNN is superior to any model other than H-ETE-GNN

1.5 Significance of Study

The study directly operationalizes the stated recommendations of Lee and Cho [9] that non-Shannon entropies are to be explored within the H-ETE-GNN framework by providing a calibrated α-grid and a regime-stratified evaluation protocol. In bridging non-Shannon entropies, we implement the information-theoretic framework of Jizba, Lavička, and Tabachová [11] on Rényi Transfer Entropy with the H-ETE-GNN model. By grounding graph construction in a tunable-order entropy, the

work supplies a principled rationale for why tail-sensitive $\alpha < 1$ should matter precisely when returns deviate from Gaussianity, and why it should be immaterial when they do not.

The study also provides a structural comparison between the topology of the networks of a Shannon-ETE and an ER-TE network, revealing whether different countries emerge as dominant volatility transmitters under tail-sensitive measurement.

Lastly, the realized-volatility forecasts are direct inputs to different finance applications like Value-at-Risk, options pricing, and dynamic hedging. If an ER-TE-based graph construction improves crisis-period RV forecasts, this could have direct implications for approaches in portfolio-level risk limits.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Theoretical Framework

2.1.1 Shannon Entropy and Transfer Entropy

Let X = $\{x_i\}$ be a discrete random variable with probability distribution P(X). The Shannon entropy of X is defined as:

$$H(X) = -\sum_{x} p(x) log_2 p(x)$$

Shannon entropy measures the average uncertainty (in bits) associated with the outcome of X. Mutual information, on the other hand, quantifies the statistical dependence of two variables. The mutual information for two random variables X and Y, denoted by I(X;Y), can be seen as follows:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(X) - H(X|Y)$$

Where H(X) is the Shannon entropy of a random variable X and H(X|Y) denotes the conditional entropy representing the uncertainty of X given Y.

The concept of Mutual Information (MI) is extended to Shannon Transfer Entropy (TE) by conditioning on X's own past [7]. TE measures the flow of information between two time series X and Y by measuring how much the past of Y reduces uncertainty about the next value of X, beyond what X's own past already provides:

$$TE_{Y\to X}^{(k,l)} = H\left(X_{t+1}|X_t^{(k)}\right) - H\left(X_{t+1}|X_t^{(k)}, Y_t^{(l)}\right)$$

$$TE_{Y\to X}^{(k,l)} = \sum_{i} p\left(x_{t+1}, x_t^{(k)}, y_t^{(l)}\right) log \frac{p\left(x_{t+1}|x_t^{(k)}, y_t^{(l)}\right)}{p\left(x_{t+1}|x_t^{(k)}\right)}$$

Where $X_t^{(k)} = \left(x_t, x_{t-1}, \ldots, x_{t-k+1}\right)$ and $Y_t^{(l)} = \left(y_t, y_{t-1}, \ldots, y_{t-l+1}\right)$ denote the past k values of X and past l values of Y. The conditional probability $p\left(x_{t+1} | x_t^{(k)}, y_t^{(l)}\right)$ measures how much better the probability of x_{t+1} is predicted when we incorporate the past information of Y along with X. The sum is then calculated over all possible joint combinations of $\left(x_{t+1}, x_t^{(k)}, y_t^{(l)}\right)$. TE is then simplified to:

$$TE_{Y \to X} = \sum_i p\left(x_{t+1}, x_t, y_t\right) log \frac{p\left(x_{t+1} | x_t, y_t\right)}{p\left(x_{t+1} | x_t\right)}$$

2.1.2 Effective Transfer Entropy

TE, although it captures nonlinear and asymmetric dependencies, can lead to a positive bias where TE cannot be negative, finite-sample noise accumulates in one direction only, leaving a strictly positive measurement even between unrelated series when estimated from finite samples. Marschinski and Kantz [10] introduced Effective Transfer Entropy (ETE) to correct this:

$$ETE(X \to Y) = TE_{observed}(X \to Y) - mean\left(TE_{shuffled}(X \to Y)\right)$$

Shuffling destroys the temporal structure of Y while preserving its marginal distribution. The $TE_{shuffled}(X \to Y)$ estimates the finite-sample bias and the statistical significance is assessed via a Z-score:

$$Z = \frac{TE_{X \to Y} - \mu_{shuffled}}{\sigma_{shuffled}}$$

Only ETE values with Z > 1.96 are retained in the adjacency matrix; all others are set to zero, yielding a sparse, directed graph.

2.1.3 Rényi Entropy

Rényi entropy [14] for order α with probability distribution P on a finite sample X is defined as:

$$H_{\alpha}(P) = \frac{1}{1-\alpha} log \sum_{x \in X} p^{\alpha}(x)$$

Rényi entropy (RE) corresponds to the average number of bits needed to optimally encode the discrete source X with probability P(X) [11]. For $\alpha > 1$, RE depends more on the probabilities of the more probable values and less on the improbable ones. Meanwhile, for $0 < \alpha < 1$, extreme events are accentuated with decreasing α. By choosing different values of α, we can focus on specific regions of the distribution, critically on the tail events that drive crisis-period dynamics in financial markets.

2.1.4 Rényi Transfer Entropy

Following Jizba, Lavička, and Tabachová [11], the Rényi Transfer Entropy of order α from Y to X is defined as:

$$T^{R}_{\alpha, Y \to X}(k, l) = H_{\alpha}\left(X_{t_{n+1}} | X_n^{(k)}\right) - H_{\alpha}\left(X_{t_{n+1}} | X_n^{(k)}, Y_n^{(l)}\right) = I_{\alpha}\left(X_{t_{n+1}} : Y_n^{(l)} | X_n^{(k)}\right)$$

Where $H_{\alpha}(X|Y)$ is the conditional entropy of order α and $I_{\alpha}(X:Y)$ is the mutual information of order α. Explicitly:

$$H_{\alpha}(X|Y) = \frac{1}{1-\alpha} log_2 \frac{\sum_{x \in X, y \in Y} p^{\alpha}(x,y)}{\sum_{y \in Y} p^{\alpha}(y)}$$

$$I_{\alpha}(X:Y) = \frac{1}{1-\alpha} log_2 \frac{\sum_{x \in X, y \in Y} p^{\alpha}(x) p^{\alpha}(y)}{\sum_{x \in X, y \in Y} p^{\alpha}(x,y)}$$

Its difference from Shannon TE lies in three components: (1) **Tail sensitivity**, the escort distribution p^{α} reweights the joint probability inside the log, allowing $\alpha < 1$ to amplify rare-event contributions. (2) **Can be negative**, unlike Shannon TE, RTE can take negative values. This occurs when knowing Y's past actually increases uncertainty about X under the Rényi measure. (3) **Shannon limit**, at $\alpha = 1$, Rényi TE reduces to Shannon TE via L'Hôpital.

2.1.5 Key Theorems

Jizba et al.'s Theorem 3.1 (Gaussian Equivalence). If the joint process $\left(X_{t'}, Y_t\right)$ is Gaussian then there is an exact equivalence between Granger Causality and RTE, namely:

$$F^{(k,l)}_{Y \to X} = 2T^{R}_{\alpha, Y \to X}(k, l)$$

Where $F^{(k,l)}_{Y \to X}$ is the standard Granger Causality measure. This result is α-independent, meaning that for Gaussian data, the choice of α is immaterial — all values give the same causal inference. This guarantees that the Rényi extension does not distort inference during calm periods when returns are approximately Gaussian [11].

Jizba et al.'s Theorem 3.2 (Heavy-tail Correction). If the joint process $\left(X_{t'}, Y_t\right)$ is α-Gaussian with $\alpha \in (\frac{1+k+l}{3+k+l}, 1]$ (i.e., finite covariance matrix region), then $F^{(k,l)}_{Y \to X} = 2T^{R}_{\alpha, Y \to X}(k, l)$ is a monotonically decreasing function of α (at fixed k and l) with zero reached at $\alpha = 1$. The leading correction is:

$$F_{Y\to X}^{(k,l)} = 2T_{\alpha,Y\to X}^{R}(k,l) + \frac{l(\alpha-1)^2}{4} + O\left((\alpha-1)^3\right)$$

This theorem is relevant to financial returns as they exhibit heavy tails, meaning they deviate from Gaussianity. For such data, Shannon TE (α = 1) and Granger Causality give the same answer, but moving α < 1 captures additional tail-driven information flow that the Gaussian Equivalence misses. The correction term $l(\alpha-1)^2/4$ quantifies exactly how much additional information is extracted [11].

2.1.6 Effective Rényi Transfer Entropy

Similar to the Shannon case, we define Effective Rényi Transfer Entropy by subtracting the RTE computed on surrogate time series:

$$ERTE_{\alpha,Y\to X} = T_{\alpha,Y\to X}^{R} - T_{\alpha,Y_{surrogate}\to X}^{R}$$

Where $Y_{surrogate}$ is a phase-randomized surrogate series that preserves the marginal distribution and autocorrelation of Y but destroys its temporal relationship with X. This construction, following Jizba et al. [11, Eq. 49], removes finite-sample bias while preserving the tail-sensitivity of the Rényi measure.

The balance of effective transfer entropy further quantifies the net direction of information flow:

$$ERTE_{\alpha,Y\to X}^{balance} = ERTE_{\alpha,Y\to X} - ERTE_{\alpha,X\to Y}$$

2.1.7 Hurst Exponent and Regime Adaptation

The Hurst Exponent H is estimated via Rescaled Range (R/S) analysis on a rolling window of length w [15]. For a time series segment T_i of length w, the Hurst Exponent is utilized to identify regime changes by analyzing the structure of the time series data. It is an effective tool to interpret the complex patterns seen in financial time series data.

Given a time series $X = \{x_1, x_2, ..., x_n\}$, we partition these into segments of length w. This results in a series of segments given by $X_w = \{T_1, T_2, ..., T_m\}$. We define each segment as $T_i = \{x_{(i-1)w+1}, x_{(i-1)w+2}, ..., x_{iw}\}$, where the total number of segments is $m = \lfloor n/w \rfloor$. By calculating the rescaled range (R/S) values across these scales, the scaling relationship of the time series can be determined. For a time series segment T_i of length w:

1. We first compute the segment mean: $\overline{T}_i = (1/w) \sum_{j=1}^{w} x_{(i-1)w+j}$
2. Compute the cumulative deviation: $Y_k^{(i)} = \sum_{j=1}^{k} x_{(i-1)w+j} - \overline{T}_i$
3. Compute the range: $R_i = max_{1 \le k \le w} Y_k^{(i)} - min_{1 \le k \le w} Y_k^{(i)}$
4. Compute the standard deviation: $S_i = \sqrt{\frac{1}{w} \sum_{j=1}^{w} (X_{(i-1)w+j} - \overline{T}_i)^2}$
5. The rescaled range R/S is given by $(\frac{R}{S})_w = \frac{1}{m} \sum_{i=1}^{m} \frac{R_i}{S_i}$

H is then estimated by regressing $log(R/S)_w$ on log w via Ordinary Least Squares (OLS).

Regime classification follows Lee and Cho (2025): using a rolling 250-day window on the MSCI World ETF (as a global proxy), a binary regime label is assigned based on whether the proportion of days with $H > 0.5$ in a trailing 10-day window exceeds a sensitivity threshold s. When regime transition is detected, the ER-TE adjacency matrix is recomputed and the GNN weights are fine-tuned (not fully retrained)

2.1.8 Graph Convolutional Networks

A graph G = (V,E) is defined by a set of nodes V and edges E, encoded in an adjacency matrix $\epsilon \mathbb{R}^{nxn}$, where A_{ij} measures the relationship from node j to node i. In this thesis, each node is one of ten country-level MSCI ETFs, and the edges are derived from the ER-TE adjacency constructed in 3.3

Classical convolutional neural networks exploit the regular grid structure of images or time series, where translation invariance allows a fixed-size kernel to be slid across the input. This assumption fails for graph-structured data, since nodes have variable neighborhood sizes and there is no canonical ordering. Graph Convolutional Networks (GCNs) generalize the convolution operation to the irregular domain of graphs, allowing each node's representation to be updated as a learned function of its own features and the features of its neighbors [16].

The spectral approach, on which Kipf and Welling [16] build, defines convolution through the graph Laplacian. For an undirected graph with adjacency A and degree matrix D, the normalized Laplacian is

$$L = I_n - D^{-1/2} A D^{-1/2}$$

L admits an eigendecomposition of L = $U\lambda U^T$, where the columns of U form a basis of graph signals analogous to the Fourier Basis on Euclidean domains. A spectral graph convolution between a signal $x \in \mathbb{R}^n$ and filter g_θ then takes the form:

$$g_\theta * x = U g_\theta(\lambda) U^T x$$

Direct computation of this convolution is expensive because it requires the full eigendecomposition of L. Defferrard, Bresson, and Vandergheynst [20] address this by approximating g_θ with a truncated expansion in Chabyshev polynomials of order K which localizes the filter to K-hop neighborhoods and removes the need for eigendecomposition.

Kipf and Welling's first-order simplification [16] further simplify this Chebyshev expansion to first order $K = 1$ and apply a renormalization trick to stabilize training yielding the layer-wise propagation rule:

$$H^{(l+1)} = \sigma(\widehat{D}^{-1/2} \widehat{A} \widehat{D}^{-1/2} H^l W^l)$$

Where $\widehat{A} = A + 1$ adds self-loops, $\widehat{D}$ is the degree matrix of $\widehat{A}$, H^l is the matrix of node features at layer l, W^l is a learnable weight matrix, and σ is a non-linear activation. Stacking l such layers allows each node to aggregate information from neighbors up to l hops away. This formulation is the spatial interpretation of a spectral GCN: at every layer, each node's representation is updated as a weighted average of its neighbors' representations, with weights determined by the (renormalized) adjacency.

2.2 Direct Predecessor - Lee and Cho (2025): H-ETE-GNN

Lee and Cho [9] introduced the Hurst-Exponent Effective Transfer Entropy Graph Neural Network (H-ETE-GNN) for one-day-ahead realized-volatility forecasting on ten country-level iShares MSCI ETFs over 2003–2024. Their hybrid model consisted of three components: (1) directed, sparse adjacency matrices built from Shannon-based Effective Transfer Entropy (ETE) with shuffled-surrogate bias correction and a $Z > 1.96$ significance threshold; (2) a Graph Convolutional Network (GCN) with multi-scale 1D temporal convolutions and three graph-convolution layers; (3) a Hurst-exponent-based regime-detection mechanism that triggers graph reconstruction and network fine-tuning when the market transitions between persistent ($H > 0.5$) and anti-persistent ($H < 0.5$) states. Against LSTM [4], GRU [19], and GNN baselines using Pearson correlation or Granger Causality [6] as edge metrics, H-ETE-GNN reports statistically significant RMSE, MAE, and QLIKE gains.

Although their hybrid model performed statistically significantly better than the aforementioned models, they identified a limitation that motivates this paper. They reported that H-ETE-GNN's relative advantage diminishes during the 2008 Global Financial Crisis, the 2020 COVID-19 shock, and the 2022 Russia–Ukraine episode, and they attribute this degradation to Shannon Entropy's uniform weighting of the probability distribution where tail events enter the ETE calculation with the same weight as routine fluctuations, despite dominating risk during crises. In their discussion of future work, they recommended exploring Rényi Transfer Entropy to capture tail-driven information flow. This paper operationalizes that recommendation through the HR-ETE-GNN model. The present paper inherits the 10-ETF universe, the 19-year sample period, the Hurst-based regime trigger, the multi-scale GCN

architecture, the MSE loss, the Adam optimizer, and the early-stopping criteria without modification; only the entropy basis of the adjacency construction is varied, so that any observed performance change can be attributed to the Rényi generalization rather than to architectural confounders.

2.3 Rényi Transfer Entropy

2.3.1 Jizba, Kleinert, and Shefaat (2012)

Jizba, Kleinert, and Shefaat [17] applied Rényi-based information transfer to financial time series. Building on Shannon Transfer Entropy of Schreiber [7], the authors introduce a Rényi-entropy generalization indexed by an order parameter $\alpha \geq 0$ and apply it to daily returns across major international stock indices. They establish empirically that α acts as a "lens" on the return distribution: small α amplifies rare tail events, while large α concentrates on the distribution mode. Shannon TE is recovered exactly as $\alpha \to 1$. Their analysis of information flow across the DJIA, NIKKEI, FTSE, and DAX shows that Rényi TE resolves multiscale and multifractal information-transfer patterns that Shannon TE cannot detect, with different α values revealing qualitatively different cross-market dependence structures.

This paper supplied the precedent that RTE with $\alpha \neq 1$ yields information beyond Shannon TE specifically on financial data, establishing that the present study's tunable-α design is not an unmotivated generalization. Second, it establishes the convention of computing RTE over an α-grid rather than a single value. However, this paper stopped at descriptive analysis and did not apply RTE for forecasting. They also did not perform any bias correction via surrogates and did not integrate RTE into

any machine learning pipeline. HR-ETE-GNN extends their work by embedding an Effective Rényi TE into a regime-adaptive forecasting model.

2.3.2 Jizba, Lavička, and Tabachová (2022)

Jizba, Lavička, and Tabachová's [11] paper on causal inference in time series via Rényi Transfer Entropy serves as one of the theoretical foundations on which this paper's identification rests. They derived Rényi Transfer Entropy (RTE) from conditional Rényi entropy and Rényi mutual information and established two results that are directly operationalized in HR-ETE-GNN.

Theorem 3.1 (Gaussian Equivalence) proves that for jointly Gaussian processes, Rényi TE equals half the Granger Causality measure for every value of α. This α-independence result is the theoretical justification for why the Rényi extension cannot distort inference during calm periods when financial returns approximate Gaussianity. The choice of α is immaterial whenever the underlying process is well-behaved.

Theorem 3.2 (Heavy-tail Correction) establishes that, for an α-Gaussian process with power-law tails, RTE acquires a correction term of leading order $l(\alpha - 1)^2/4$ on top of the Shannon-limit value, quantifying how much additional tail-driven information is extracted by moving below the Shannon limit.

The two theorems jointly deliver a principled argument for why tail-emphasizing $\alpha < 1$ should matter precisely when returns deviate from Gaussianity (crisis) and should not matter when they do not (calm periods). They also define the Effective Rényi TE via phase-randomized surrogates; HR-ETE-GNN adopts this construction verbatim for bias correction.

2.4 Graph Neural Network For Financial Networking

2.4.1 Kipf and Welling (2017) - Graph Convolutional Network

Kipf and Welling (2017) [16], introduced the Graph Convolutional Network (GCN) which is a first-order spectral graph-convolution layer in which each node aggregates a linear combination of its neighbors' features at every layer, followed by a pointwise non-linearity. The propagation rule $H^{(l+1)} = \sigma(\widehat{D}^{-1/2}\widehat{A}\,\widehat{D}^{-1/2}H^{l}\,W^{l})$, with $\widehat{A} = A + 1$ and $\widehat{D}$ the corresponding degree matrix, reduces graph learning to sparse matrix multiplications tractable at scale. Although the original application is semi-supervised node classification on citation graphs, the formulation has become the default choice wherever practitioners need a simple, well-understood message-parsing layer over a fixed adjacency.

Lee and Cho (2025) [9] adopt a stack of three GCN layers for their financial application, and HR-ETE-GNN inherits this architectural choice unchanged in order to preserve the clean-ablation protocol. The GCN's requirement of a pre-specificed adjacency matrix is precisely what makes the entropy-basis intervention explored in this paper meaningful: the learnable parameters of the network operate on whatever graph is supplied, so replacing Shannon-ETE edges with ER-TE edges isolates the contribution of the edge-construction metric

CHAPTER III

METHODOLOGY

To compare the difference between the application of Rényi Transfer Entropy and Shannon Transfer Entropy to predict stock volatility, our methodology follows Lee and Cho's [9] hybrid model.

3. Methodology as Replicable

3.1 Data

This paper will use daily closing prices of ten MSCI country-level iShares ETFs representing both developed and emerging markets: (1) EWA, iShares MSCI Australia, (2) EWC, iShares MSCI Canada, (3) EWG, iShares MSCI Germany, (4) EWJ, iShares MSCI Japan, (5) EWT, iShares MSCI Taiwan, (6) EWU, iShares MSCI United Kingdom, (7) EWW, iShares MSCI Mexico, (8) EWY, iShares MSCI South Korea, (9) EWZ, iShares MSCI Brazil, (10) EZA, iShares MSCI South Africa. Additionally, the iShares MSCI World ETF is used for the Hurst-exponent-based regime detection. The data gathered would be from February 7, 2003, to May 29, 2024.

Given p_t as the current closing price at time t of an ETF and p_{t-1} as the closing price of the day before, we compute the return r_t at time t as the following:

$$r_t = ln\left(p_t / p_{t-1}\right)$$

Realized volatility is computed from the past n trading days of log-returns:

$$RV_t = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(100 r_{t-i}\right)^2}$$

Where t refers to the specific date, and n denotes the number of past observations. For this study, we will compute realized volatility using the past 20 days of log-returns.

The supervised samples are formed with a look-back window of 20 trading days and split chronologically into training (60%), validation (20%), and test (20%) blocks. All graph estimation, hyperparameter selection, and Rényi-order selection use the training and validation blocks only; the ER-TE adjacency matrix is estimated on training-block returns alone, so that no-test period observation informs the graph the model uses to forecast it.

3.2 Crisis sub-samples

Three crisis periods are defined for regime-stratified evaluation: (1) the Global Financial Crisis (GFC), spanning September 2008 to March 2009; (2) COVID-19, spanning February 2020 to June 2020; (3) the Russia–Ukraine conflict, spanning February 2022 to May 2022.

These windows are defined by reference to the documented episodes of extreme market stress themselves, and not by the behaviour of the Hurst Exponent.

3.3 ER-TE Estimation Procedure

Step 1 — Rényi entropy estimation. We utilize the LPS estimator of Leonenko, Pronzato, and Savani [18], a k-nearest-neighbor (k-NN) estimator of Rényi entropy, for a sample $\{x_1, x_2, \ldots, x_N\}$ in d dimensions for order $\alpha \neq 1$:

$$\hat{H}_\alpha = \frac{1}{1-\alpha} log_2 \left[C_k \cdot (N-1)^{1-\alpha} \cdot B_d^{1-\alpha} \cdot \frac{1}{N} \sum_{i=1}^{N} \rho_i^{d(1-\alpha)} \right]$$

Where ρ_i is the distance from x_i to its k-th nearest neighbor and B_d is the volume of the d-dimensional unit ball.

Step 2 — Conditional Rényi entropy. The conditional Rényi entropies $H_\alpha\left(X_{t+1}|X_n^{(k)}\right)$ and $H_\alpha\left(X_{t+1}|X_n^{(k)}, Y_n^{(l)}\right)$ are estimated using the same k-NN approach in the appropriate joint spaces (dimensions d = k + 1 and d = k + l + 1, respectively).

Step 3 — RTE computation. For each ordered pair (Y, X) of ETFs:

$$T^R_{\alpha,Y\to X} = H_\alpha\left(X_{t+1}|X_n^{(k)}\right) - H_\alpha\left(X_{t+1}|X_n^{(k)}, Y_n^{(l)}\right)$$

with lag parameters k = 1, following Lee and Cho [9].

Step 4 — Effective correction. For each pair, generate m shuffled surrogates of Y and compute:

$$ERTE_{\alpha,Y\to X} = T^R_{\alpha,Y\to X} - \frac{1}{m}\sum_{i=1}^{m} T^R_{\alpha,Y_i\to X}$$

Step 5 — Significance testing. Compute the Z-score against the shuffled distribution. Retain only ER-TE values with Z > 1.96 in the adjacency matrix; set all others to zero.

Step 6 — Adjacency matrix. The resulting n × n sparse, directed, weighted matrix A_α serves as the GCN adjacency for a given training window and Rényi order α. Rows of A_α are normalised to sum to one before use.

Row normalisation is a departure from the specification of Lee and Cho [9]. And it is made for experimental control rather than in spite of it. Under their node-update rule (Section 3.4) the neighborhood term enters in proportion to the raw magnitude of the

adjacency entries, so the influence of the graph on the forecast is a function of the scale of the edge weights and not only of their pattern. Rényi and Shannon transfer entropies do not share a common scale: the order α enters the Rényi entropy through a factor $1/(1-\alpha)$, so ER-TE magnitudes vary systemically with α even where the underlying network is changed. On raw weights, a comparison between $\alpha = 1$ and $\alpha = \alpha^{*}$ arms would therefore confound a difference in edge topology, which is the object of this study. Row Normalisation removes that confound by placing every adjacency on a common scale, and it is applied identically to both arms and to every secondary graph model.

Step 7 — Network Comparison. For each matrix, each ETF's out-degree is computed as the number of retained outgoing edges and its weighted out-degree as the sum of their ER-TE values. The two networks are then compared on three quantities: (i) the identity and ordering of the top-K transmitters, K = 3; (ii) the number of directed edges common to both matrices, reported as an overlap table alongside adjacency heatmaps; and (iii) Kendall's rank correlation τ [22] between the two out-degree rankings, with a 90% confidence interval obtained by stationary bootstrap [21] with expected block length 20 trading days. All three are reported separately for crisis and non-crisis windows.

The confidence interval on τ is reported to convey how much of the observed rank divergence is compatible with sampling variation. It is a single pre-specified quantity rather than a battery of edge-level tests, and it is presented as a descriptive aid; the language used to discuss it in Chapters IV and V is that of compatibility with noise, not of statistical significance.

3.4 Graph Convolutional Network

The GNN architecture follows Lee and Cho [9] and is grounded in the spectral GCN framework of Kipf and Welling [16], to enable a clean ablation.

Multi-scale temporal feature extraction. Given an input time series of length t for each of n ETFs, multi-scale features are extracted via 1D convolutional filters with kernel sizes $k_i \in \{3, 5, 7\}$, each with 12 channels:

$$f = \oplus_{i=0}^{q} ReLU\left(k_i \cdot v + b_i\right)$$

Where $\oplus$ denotes concatenation across filter outputs. The result is a feature matrix $H \in R^{n \times d}$.

Message passing. The adjacency matrix A is the ER-TE matrix. The GCN update rule for node i at layer l + 1 is:

$$h_i^{l+1} = ReLU\left(h_i^l W_1^l + \sum_{j \in N(i)} A_{ji} h_j^l W_2^l\right), \quad h^0 = f$$

Where W_1^l and W_2^l are learnable weight matrices, A_{ji} is the ER-TE weight of the directed edge from node j to node i, and N(i) is the set of neighbors of node i. With three GCN layers, each node aggregates information from neighbors up to three hops away.

Output. The final GCN layer produces a prediction vector of dimension (1 × n), representing the one-day-ahead realized-volatility forecast for each of the **n** ETFs.

3.5 Hyperparameter Configuration

The Rényi order α is introduced as an additional hyperparameter alongside those defined in Lee and Cho [9]. All hyperparameters are tuned via Bayesian Optimization.

Table 3.5.1 Hyperparameters

Hyperparameter	Search Space	Description
Learning Rate	$[1\,x\,10^{-2} - 1\,x\,10^{-5}]$	Adam optimizer step size
Batch Size	[4, 8 , 16]	Mini-batch size
Hidden State Size (Layer 1)	[10-100]	GCN hidden dimension
Hidden State Size (Layer 2)	[10-100]	GCN hidden dimension
Input Size (M)	[20,40,60]	Look-back window in days
Training Period (N)	[250, 500, 750]	Training window length
Regime Sensitivity (s)	[6 , 7, 8]	Hurst threshold for regime change
Shuffle Simulations (m)	[100]	Number of surrogate series for ETE
Rényi Order (α)	{0.3, 0.5, 0.7, 0.9, 1.0, 1.5, 2.0}	Tail-sensitivity parameter

When $\alpha = 1.0$, HR-ETE-GNN reduces exactly to the original H-ETE-GNN [9], ensuring backward compatibility.

3.6 Regime-Adaptive Retraining Protocol

Following Lee and Cho [9]:

- Compute the daily Hurst exponent H(t) on the MSCI World ETF using a 250-day rolling R/S window.
- For each day t, count the number of days in $\{t - 9, \ldots, t\}$ where $H > 0.5$.
- If this count exceeds the threshold s, classify as "persistent" regime; otherwise, "anti-persistent."
- When the regime label changes from the previous day, trigger:
    - Recomputation of the ER-TE adjacency matrix on the most recent N days.
    - Fine-tuning of the GNN weights.
- If the regime remains unchanged, the current model and adjacency are used for prediction without modification.

3.7 Benchmark Models

The primary comparison is HR-ETE-GNN vs H-ETE-GNN. Secondary comparisons against other models provide context.

Table 3.7.1 Baseline Models and their Characteristics

Model	Edge Metric	GNN	Regime Adapt	α
HAR-RV	N/A (univariate)	No	No	N/A

LSTM [4]	N/A (univariate)	No	Yes	N/A
GRU [19]	N/A (univariate)	No	Yes	N/A
GNN-Pearson	Pearson correlation	GCN [16]	No	N/A
GNN-Granger	Granger causality [6]	GCN [16]	No	N/A
GNN-TE	Shannon TE [7]	GCN [16]	No	N/A
GNN-ETE	Shannon ETE [10]	GCN [16]	No	N/A
H-ETE-GNN [9]	Shannon ETE	GCN	Hurst	$\alpha = 1$
HR-ETE-GNN (this work)	Rényi ETE	GCN	Hurst	Tuned

3.8 Evaluation Metrics

The pre-registered primary loss function is QLIKE, evaluated on the variance scale. Given the realized volatility variance σ^2, and the model's forecast variance $\widehat{\sigma}^2$ of ETF i on day t, we compute QLIKE as:

$$r_{t,i} = \frac{\sigma^2_{t,i}}{\widehat{\sigma}^2_{t,i}}$$

$$QLIKE_{t,i} = r_{t,i} - logr_{t,i} - 1$$

All tests in Section 3.9 operate on a single daily series. For each test day t, and ETF i, the loss differential is:

$$d_t = \frac{1}{n} \sum_i [QLIKE_{t,i}(H - ETE - GNN) - QLIKE_{t,i}(HR - ETE - GNN)]$$

that is, the cross-sectional average across the n = 10 ETFs of the per-asset loss difference, taken first, so that each test is applied to T daily observations rather than to a pooled panel of n × T. A positive d_t means HR-ETE-GNN incurred the smaller loss on day t, and every hypothesis below is therefore one-sided in the direction d > 0. Pooling the panel would be inappropriate: the ten ETFs' losses are strongly correlated within a date, a time-series HAC estimator does not correct for that cross-sectional dependence, and a pooled test would consequently understate the standard error and overstate significance.

All standard errors are heteroskedasticity- and autocorrelation-consistent, computed with the Newey–West estimator [25] under a Bartlett kernel with bandwidth $\left[4(T/100)^{(2/9)}\right]$, where T is the number of test days; the bandwidth actually used is reported. HAC correction is necessary rather than precautionary here, since volatility clusters and loss differentials are in consequence strongly serially correlated, so that an variance estimate would manufacture significance.

3.9 Hypotheses and Test Procedure

3.9.1 Hypothesis 1a — Average predictive accuracy

H_0: Over the test period, HR-ETE-GNN and H-ETE-GNN have equal expected loss, and any observed difference is sampling variation.

H_1: HR-ETE-GNN has strictly lower expected loss than H-ETE-GNN.

To test for this we use The Giacomini–White test of conditional predictive ability [26] in its unconditional form, one-sided, with the HAC standard error. The statistic is $t = \bar{d} / SE(\bar{d})$ and the one-sided p-value is $1 - \Phi(t)$.

H_0 is rejected at the 5% level for $t > 1.645$. In addition to the p-value, the mean differential $\bar{d}$, its expression as a percentage of the baseline's mean loss, and a one-sided 95% lower confidence bound are reported. A statistically significant improvement smaller than 5% of baseline loss is reported as statistically detectable but economically immaterial. A negative $\bar{d}$ with a large statistic is reported as evidence that the Shannon baseline is superior; the test is not converted to a two-sided form after the fact.

3.9.2 Research Question 2 — Descriptive network comparison

Research Question 2 concerns the structure of the estimated networks rather than the accuracy of the forecasts they support, and it is addressed descriptively. No hypothesis test is attached to it, and no claim of statistical significance is made about any structural difference reported under it. Its findings are secondary to those of Section 3.9.1

During crisis windows, the set of dominant volatility-information transmitters, the top-ranked source ETFs by out-degree in the retained adjacency matrix, is expected to differ between Shannon ($\alpha = 1$) and tail-sensitive (α^*) measurement, and to coincide outside them. The basis for this expectation is the theory of Section 2.1.5 and is directional: by Theorem 3.1 the two measures agree exactly when returns are Gaussian, which is the calm-period case, and by Theorem 3.2 they diverge as returns become heavy-tailed.

The three quantities of Section 3.3, Step 7, top-K transmitter identity, edge overlap counts with adjacency heatmaps, and Kendall's τ between the two out-degree rankings with a stationary-bootstrap confidence interval, computed separately for crisis and non-crisis windows.

A formal test of the null that the two transmitter rankings are identical would require a resampling procedure over the edge set together with correction for the ninety ordered pairs screened simultaneously. That machinery lies outside the scope defined in Section 1.4. The consequence is stated plainly rather than left to the reader to infer: the structural differences reported in Chapter IV may in part reflect estimation noise, and no significance is claimed for them. They are offered as a characterisation of what the two measures see, and as context for the forecasting results, which are where the inferential weight of this study rests.

Relation to the forecasting hypotheses. The structural comparison is a claim about measurement, not about prediction, and the two are independent by construction. The two entropy bases may describe visibly different networks without that difference translating into forecast accuracy under the architecture of Section 3.4, and conversely. The two sets of results are therefore reported without conditioning either

on the other, and in particular a divergence in network structure is not offered as evidence for, nor its absence as evidence against, Hypotheses 1a.

CHAPTER IV

PRELIMINARY RESULTS

4.1.1 Preliminary Results

The results in this section come from a reduced-scope execution of the pipeline described in Chapter III and are reported to demonstrate the evaluation format rather than to establish a finding. They should not be read as estimates of the confirmatory result.

Table 4.1.1 Mean QLIKE by Model

Model	QLIKE	vs. baseline
H-ETE-GNN	0.01360	–
HR-ETE-GNN	0.01430	-5.15%

Table 4.1.2 Predictive Accuracy Comparison

Quantity	Value
Test Days (T)	314
Mean loss differential ($\bar{d}$)	-0.0007
Relative Change vs Baseline	-5.15%
HAC Standard Error	0.000363
GW t-statistic	-1.928
One-sided p-value	0.973
Decision at 5%	Failed to reject

Over T = 314 test days, mean QLIKE was 0.01360 for H-ETE-GNN and 0.01430 for HR-ETE-GNN at $\alpha = 0.5$, giving a mean loss differential of $\bar{d} = -0.000700$, or −5.15% of baseline loss. The HAC standard error was 0.000363, yielding a Giacomini–White statistic of $t = -1.928$ and a one-sided p-value of 0.973 against the pre-registered alternative. The null of equal predictive accuracy is not rejected at the 5% level. Under the decision rule of Section 3.9.1 a negative differential of this magnitude is reported as favouring the Shannon baseline rather than as an absence of difference; consistent with that rule, the test is not restated in the opposite direction.

Table 4.1.3 Out-Degree by Country, per α

Country	$\alpha = 0.5$ (Rényi)	$\alpha = 1$ (Shannon)
Germany	5	6
Australia	2	0
Canada	3	1
UK	3	3
South Korea	2	1
Brazil	0	0
Japan	1	1
Taiwan	2	4
Mexico	1	1
South Africa	1	3

Germany is the leading volatility-information transmitter under both measures, with out-degree 6 under Shannon and 5 under Rényi. The top 3 transmitters for Rényi are Germany, Canada, and the UK. Meanwhile, for Shannon they are Germany , Taiwan , and the UK.

CHAPTER V

CONCLUSION

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APPENDIX